University of Prishtina “Hasan Prishtina”

Faculty of Electrical and Computer Engineering

Green Team Documentation

Team 9: Bleta Punëtore

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Table 1: Public Services of Team 9 7

Introduction

This document presents a step-by-step explanation on how users can access the network and its features. Moreover, it gives an insight into how the system and servers operate and interact, and as such gives light to the internal workings of the network infrastructure.

Following the steps provided in this document, users can log in and begin accessing the resources they need within the organization with ease, as far as their specific permissions and responsibilities allow.

1. Public Services

The publics services that are accessible for the users and their domain and server names are mentioned on Table 1:

Domain Name	Server Name
ad.team9.isucdc.com	Active Directory
tickets.team9.isucdc.com	Ticketing Machine
mgmt.team9.isucdc.com	Management Backend
admin.team9.isucdc.com	Admin User Interface
www.team9.isucdc.com	Public Website

Table 1: Public Services of Team 9

2. Servers

This section provides a comprehensive overview of each server's function and how access is tailored to each user role

2.1 Active Directory (AD)

Active Directory plays a crucial role in our server infrastructure by centralizing user account management, security settings, and access permissions across the network. It acts as the backbone of authentication and authorization, ensuring users can securely access the systems and services they need.

By keeping everything organized and secure, Active Directory simplifies user management, streamlines access control, and strengthens overall network security. It helps ensure a smooth and efficient workflow, making it easier to manage accounts while maintaining a structured and protected IT environment.

2.1.1 Remote Desktop connection

This is for the IT Administrators only.

Remote Desktop Protocol (RDP) is a network communication protocol that enables users to remotely access and control a computer as if they were physically present. It facilitates secure management of files, applications, and system settings over a network connection, making it a valuable tool for remote administration and support.[1] (Deland-Han, 2025)

1. Open the remote desktop connection service and fill in the computer and username fields and click connect as shown in Figure 1

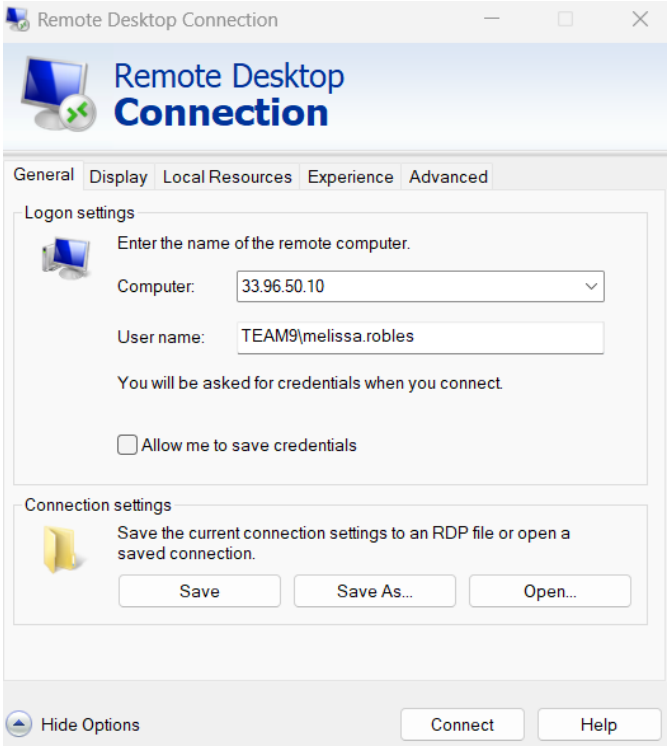

Figure 1 RDP Window

2. Afterwards, give your credentials as demonstrated in Figure 2

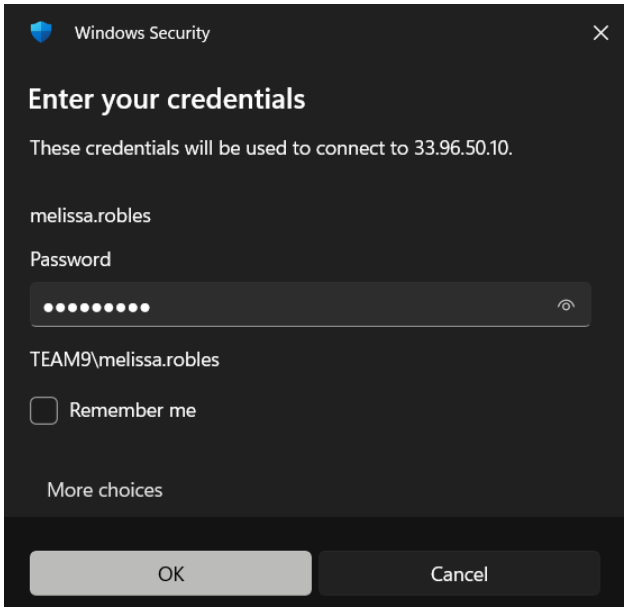

Figure 2 Providing Credentials for user 'meliss.robles'

2.1. If a pop-up window like Figure 3 comes up, click yes to continue

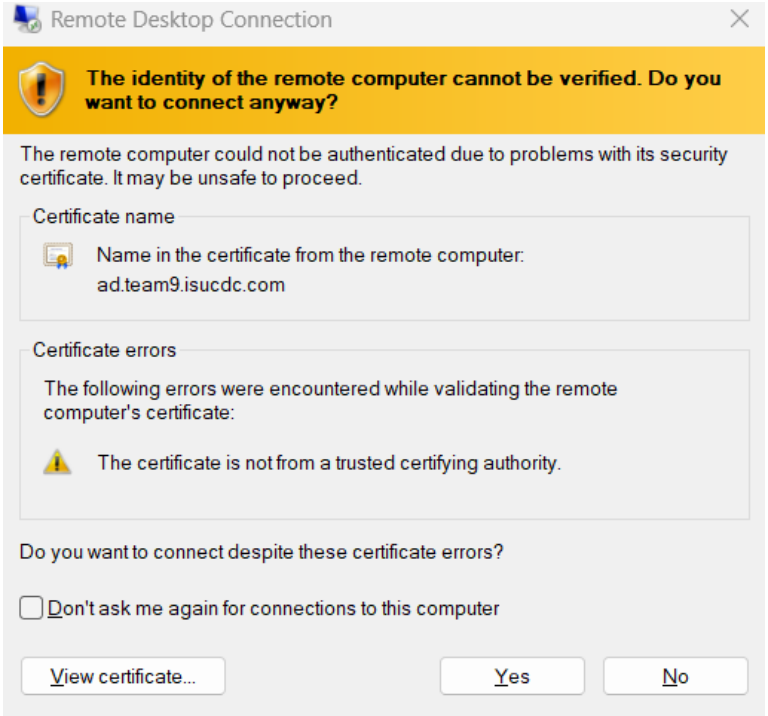

Figure 3 RDP Security Pop Up

3. You are now connected to AD as illustrated in Figure 4, when all tasks are done click on the 'x' to close the connection.

Figure 4 RDP connection has beem established

2.2 Ticketing Machine

The ticketing machine serves as the core application server for the city's ticketing infrastructure, hosting the backend logic responsible for issuing, validating, and managing transportation tickets. It processes ticket generation requests, verifies ticket authenticity, enforces validation rules, and maintains transactional records. In essence, it acts as the central authority that determines whether a ticket is valid, expired, duplicated, or fraudulent, ensuring consistency across all connected ticket readers and client interfaces.

2.2.1 Remote Desktop Connection

This is for the IT Administrators and Ticket Maintenance only.

1. Open the remote desktop connection service and fill in the computer and username fields and click connect as shown in Figure 5

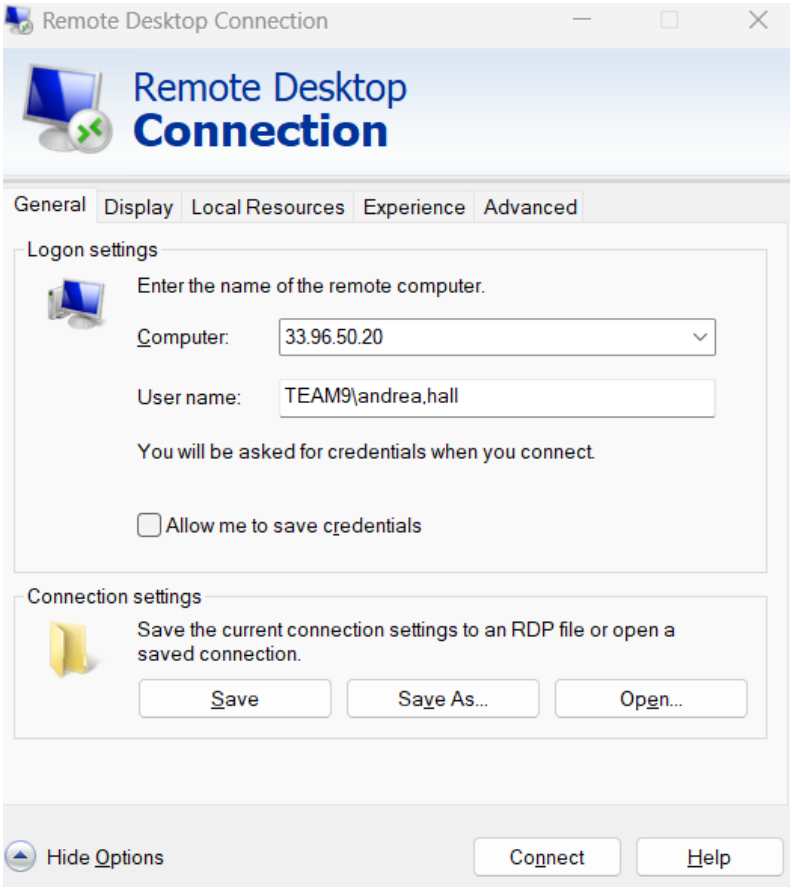

Figure 5 RDP window

2. Afterwards, you give your credentials as demonstrated in Figure 6

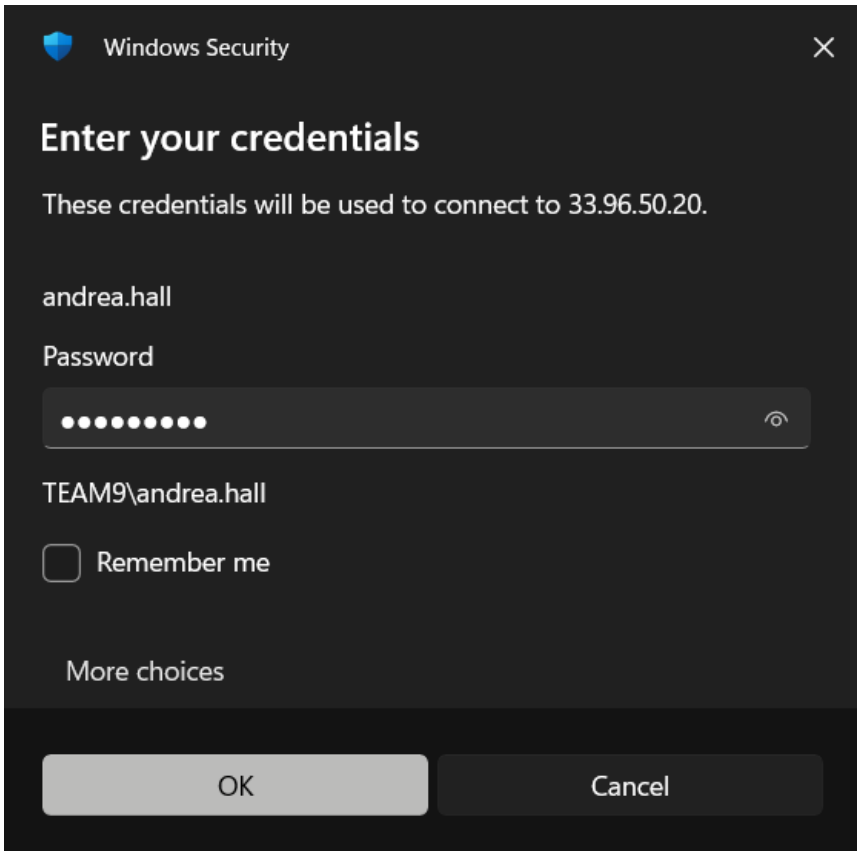

Figure 6 Providing credentials for user 'andrea.hall"

2.1. If a pop-up window like Figure 7 comes up, click yes to continue.

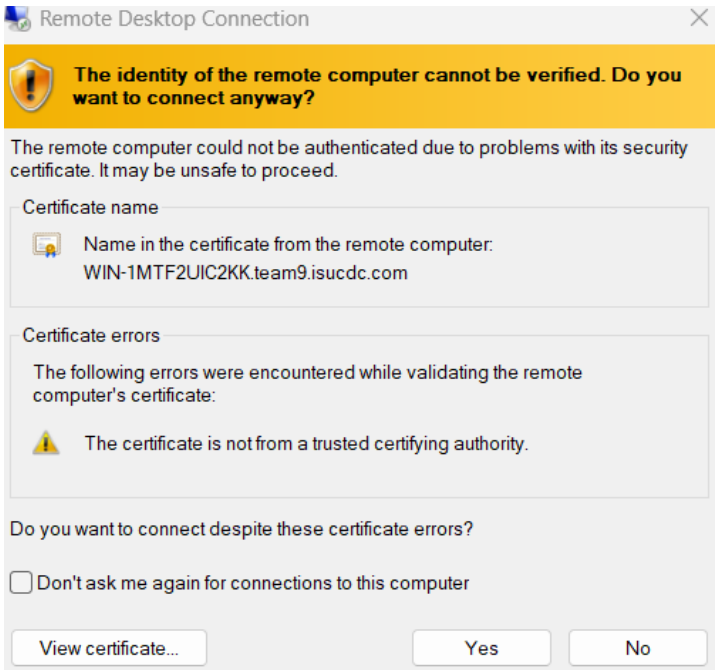

Figure 7 RDP Security Pop Up

3. You are now connected to Windows client as illustrated in Figure 8, when all tasks are done click on the 'x' to close the connection.

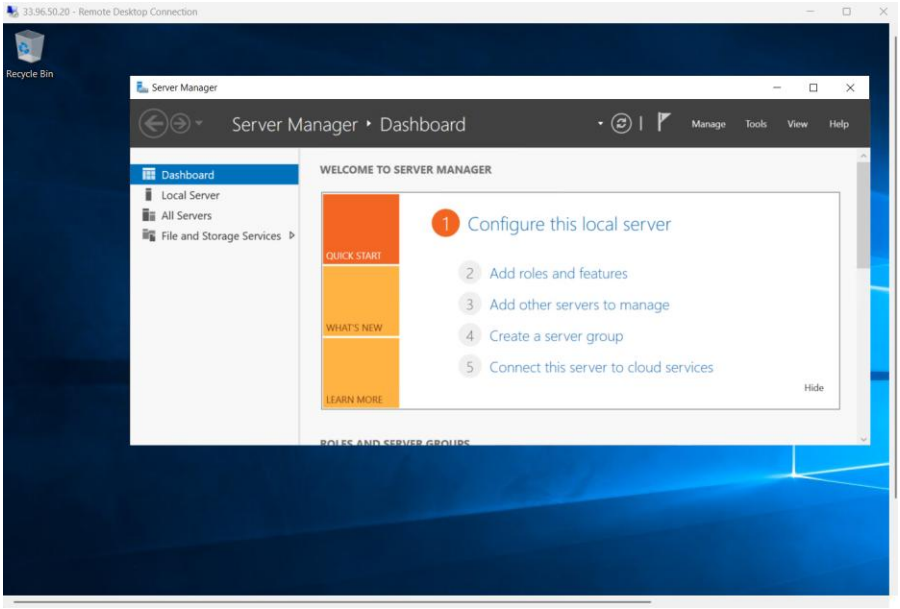

Figure 8 RDP Connection has been established

2.2.2 Ticketing Machine Health Check and Help API Calls

1.Enter the tickets.team9.isucdc.com/api/tickets/health URL in the Browser, as shown in Figure 9

Figure 9 Navigating to the health endpoint of the Ticketing Systems Api

2. See that it returns 200 Ok response like figure 10.

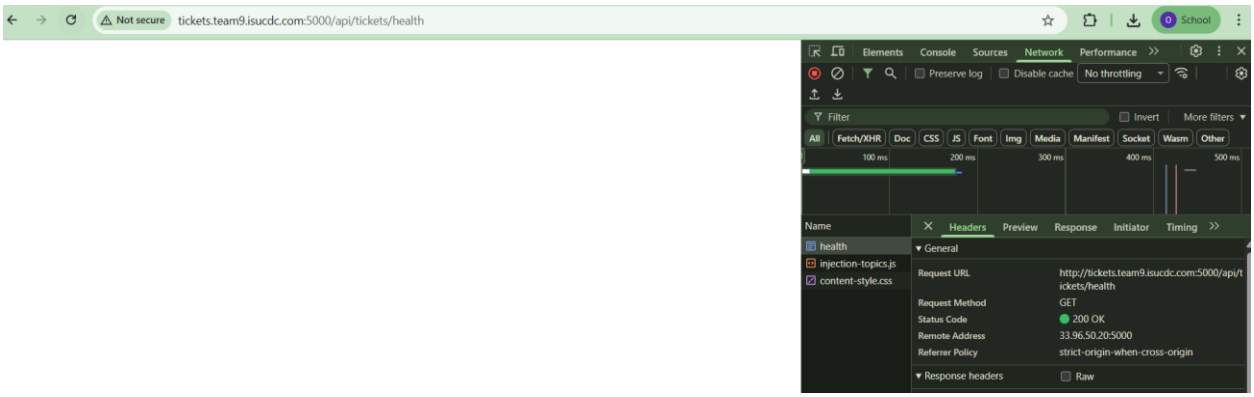

Figure 10 The endpoint returns a 200 OK status code, the API is Active

2.3 Management Backend Server

This machine functions as the central transit management backend for the city, supporting all core operational services across the transportation network. It handles user authentication, processes ticket payments, manages account data, and coordinates fleet and route configurations. By centralizing these responsibilities, it ensures that all administrative and customer-facing systems operate using consistent, validated data and business logic.

2.3.1 Logging in via SSH

The following information applies only to IT Administrators.

1. A secure connection to the Management server can be established by the SSH command with your username, e.g "username@mgmt.team9.isucdc.com". An example is shown in Figure 11.

```
C:\Users\olt_v>ssh melissa.robles@mgmt.team9.isucdc.com
```

Figure 11 SSH command for user "melissa.robles"

2. After executing the command, the user will be prompted to provide a password. Upon the event of the successful authentication, the user will get a confirmation message as shown in Figure 12. You are now logged into your account and can continue with your tasks.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'mgmt.team9.isucdc.com' (ED25519) to the list of known hosts.
Authorized users only. All activity is monitored and logged.
melissa.robles@mgmt.team9.isucdc.com's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Feb 19 09:25:05 2026 from 68.32.254.50
[melissa.robles@mgmt ~]$
```

Figure 12 Successful Login

3. Whenever the user is ready to logout from the session and wants to terminate the connection, they just need to enter 'exit' or 'logout' which are shown in figures 13 and 14 respectively.

```
[melissa.robles@mgmt ~]$ exit
logout
Connection to mgmt.team9.isucdc.com closed.
```

Figure 13 Terminating SSH connection via 'exit' command

```
Last login: Fri Feb 20 21:08:01 2026 from 199.100.16.101
[melissa.robles@mgmt ~]$ logout
Connection to mgmt.team9.isucdc.com closed.
```

Figure 14 Terminating SSH connection via 'logout' command

2.4 Admin User Interface Server

This machine hosts the city's administrative portal, providing centralized visibility into the operational status of the transportation system. Through this interface, authorized personnel can monitor vehicle activity, review system metrics, and oversee the overall performance of transit services. It serves as the primary management console for real-time operational awareness.

2.4.1 Logging in via RDP

The following information applies only to IT Administrators.

1. Open the remote desktop connection service and fill in the computer and username fields and click connect as shown in Figure 15.

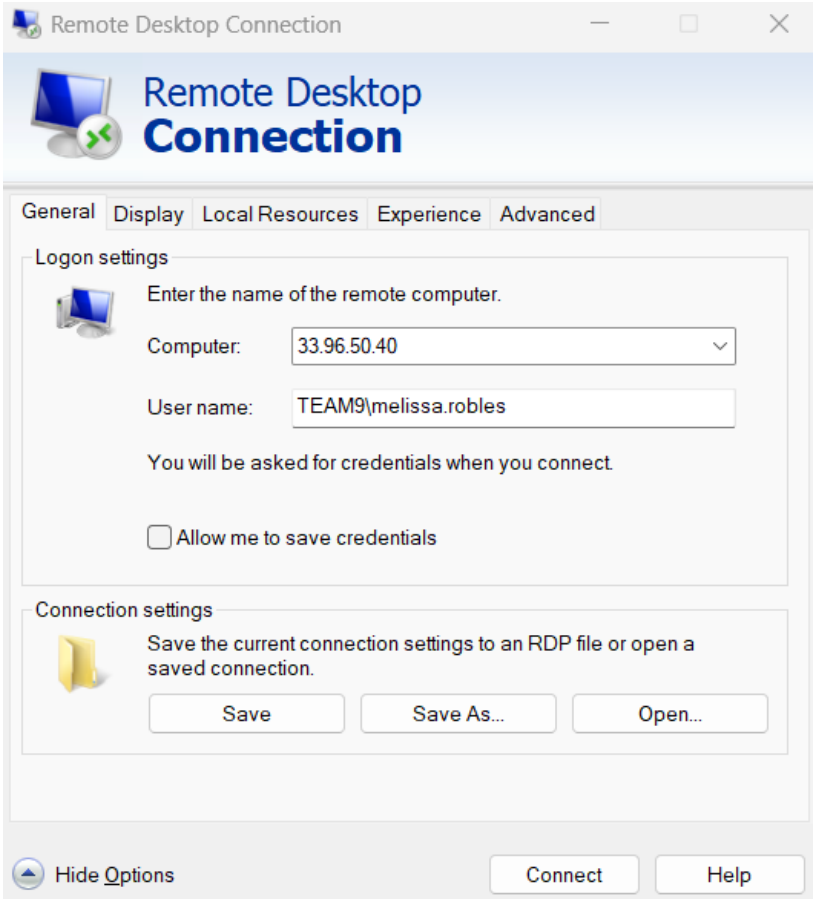

Figure 15 RDP window

2. Afterwards, you give your credentials as demonstrated in Figure 16.

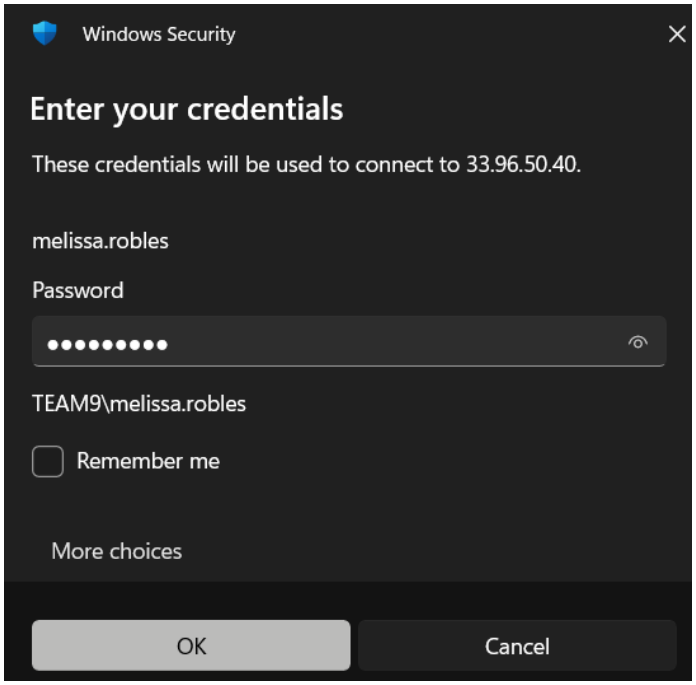

Figure 16 Providing credentials for user 'melissa.robles'

2.1. If a pop-up window like Figure 17 comes up, click yes to continue.

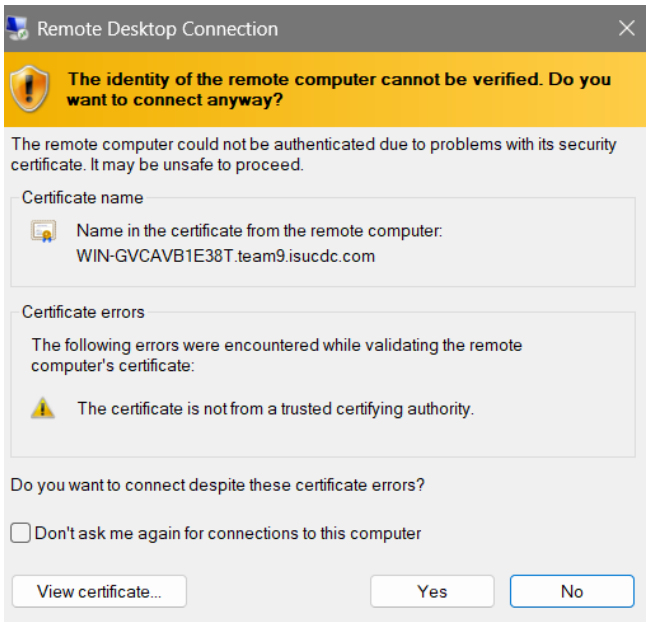

Figure 17 RDP Security Pop Up

3. You are now connected to Windows client as illustrated in Figure 18, when all tasks are done click on the 'x' to close the connection.

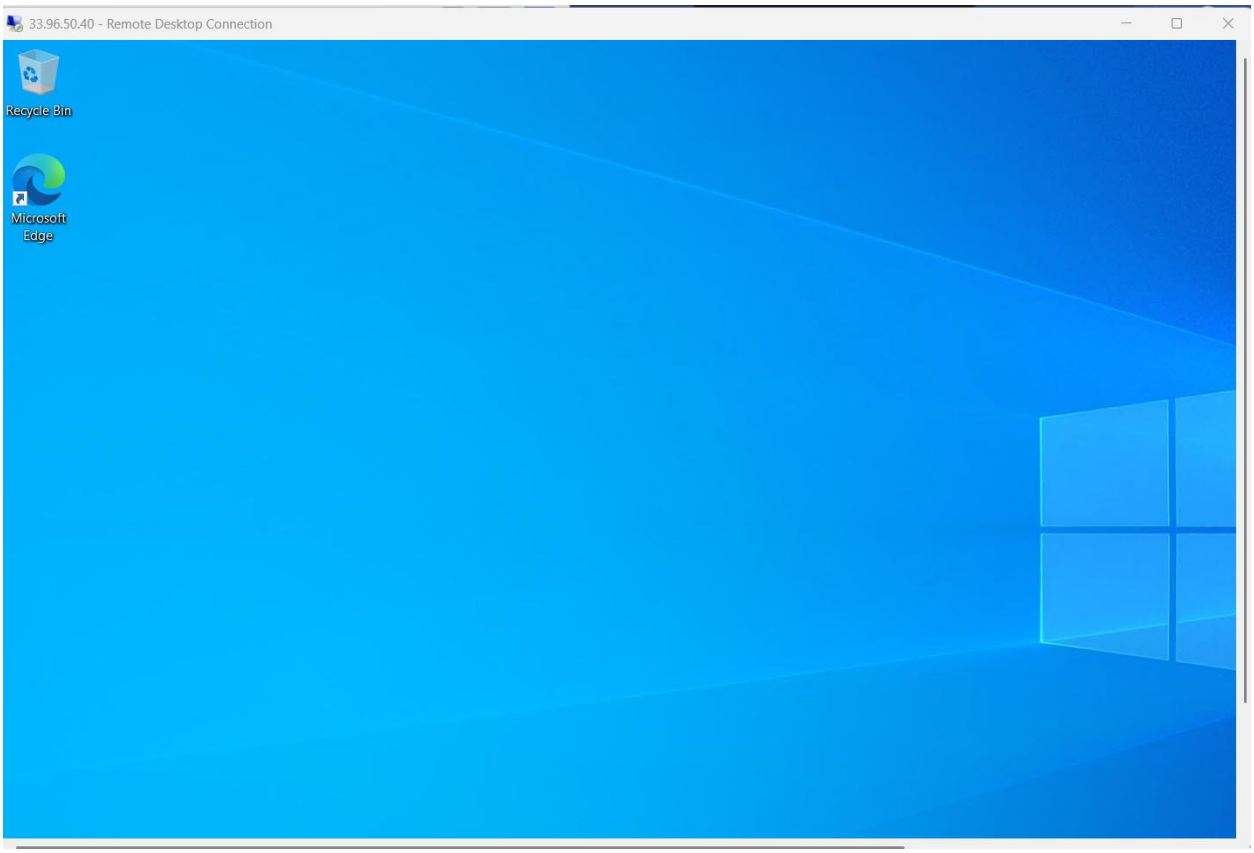

Figure 18 RDP connection has been established

2.4.2 Logging in Procedure for the Admin UI

This applies only to Transit Administrators.

1. Navigate to admin.team9.isucdc.com in the Browser it will redirect users to the login page, as demonstrated in Figure 19

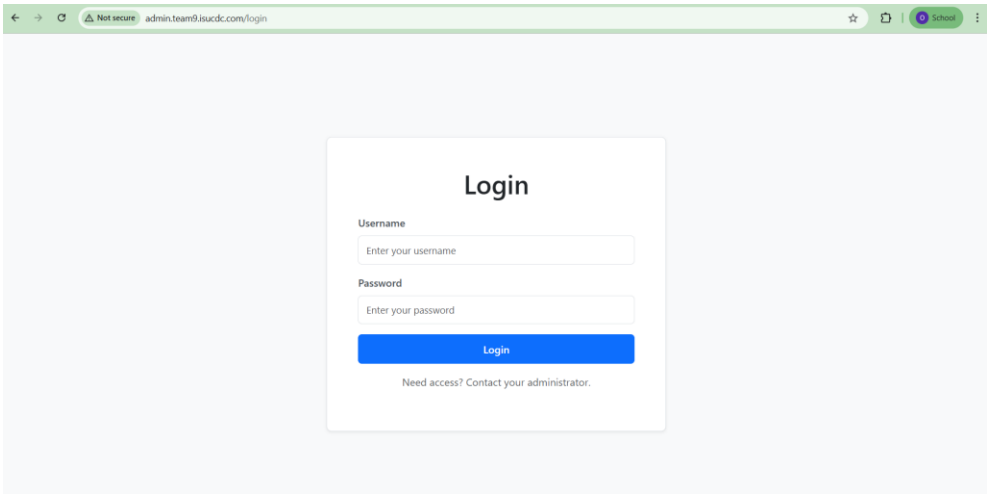

Figure 19 The Login Page of the Admin UI

2. After logging in they will be redirected to home page, as shown on Figure 20

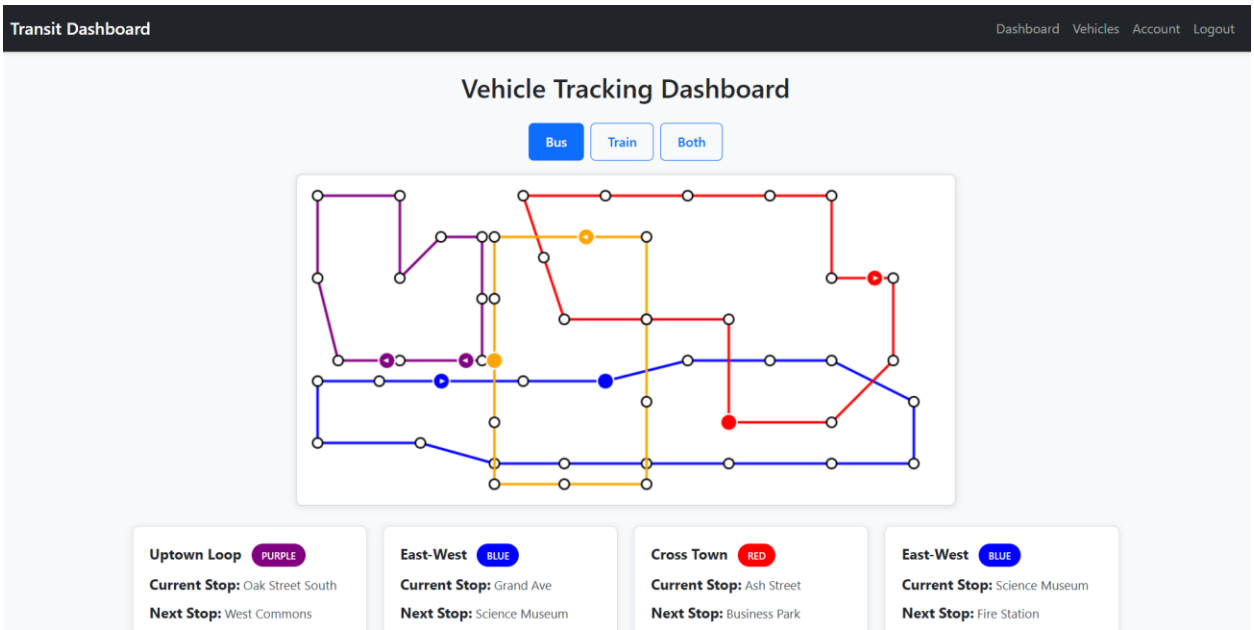

Figure 20 Dashboard after Successful Login

2.4.3 Vehicle status and capacity modification

This applies only to Transit Administrators

1. Log into the web page

2. Navigate to the /vehicles page by clicking on the Vehicles option on the navbar. After this the users will land on the vehicle page, as illustrated in Figure 21

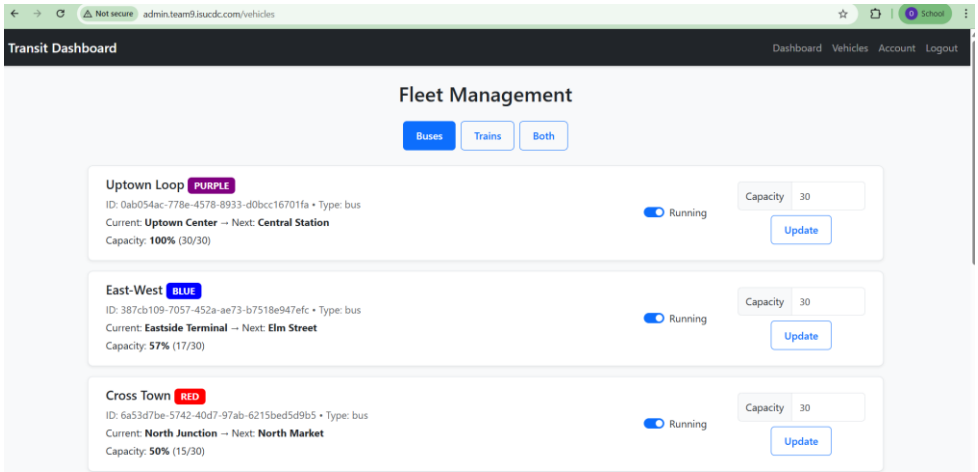

Figure 21 The Vehicle page

3. Do any modifications needed to the capacity or status and click Update. After the action has been completed an informational pop up will appear, as shown in Figure 22

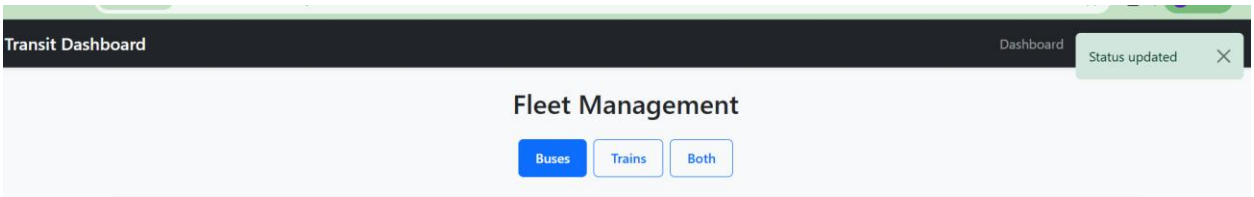

Figure 22 The pop up on the top right indicating that the status of a vehicle has been changed.

2.4.4 Password Change Procedure

This applies only to Transit Administrators

1. Log into the web page

2. Navigate to the /account page by clicking on the Vehicles option on the navbar. After this the users will land on the vehicle page, as illustrated in Figure we

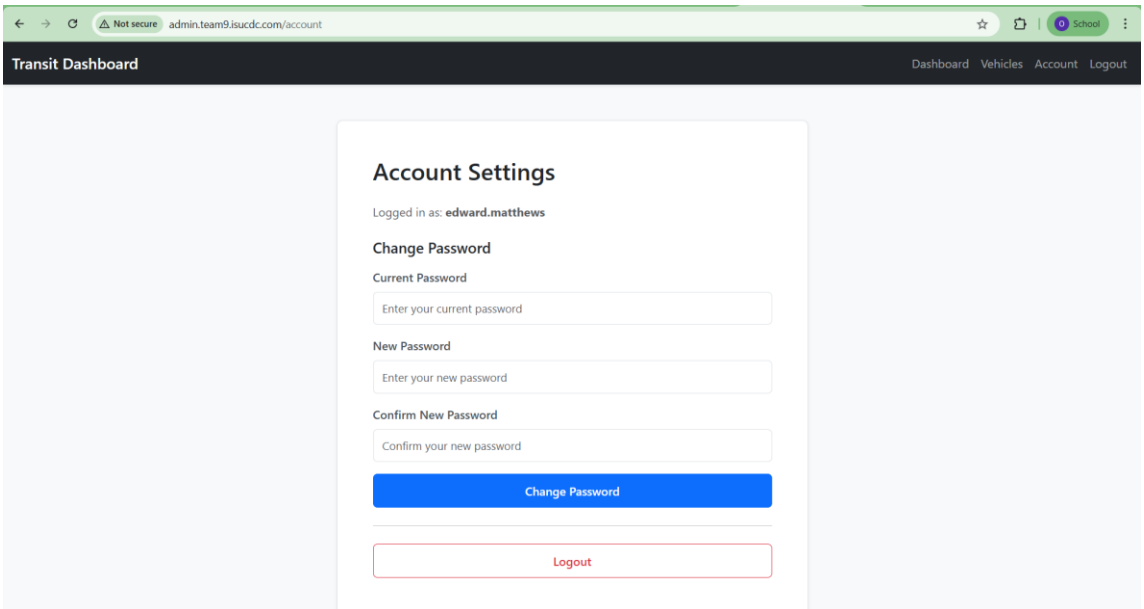

Figure 23 Account Page

3. Fill in all the required fields and click on "Change Password" button and observe that the password for the user has been changed.

Note:

The authentication and vehicle endpoints communicate with the Management Backend Server. So, every Api function of the admin UI page will function correctly if the Management Backend server is active

2.5 Public Website Server

This machine hosts the city's public-facing transportation website, serving as the primary digital access point for citizens. Through this platform, users can create accounts, authenticate, browse bus and train routes, manage personal information, and purchase tickets. It provides a centralized interface that connects the public directly to the city's transit services.

Within the overall infrastructure, this system functions as the front-end layer that communicates with the management backend services to process transactions, validate tickets, and retrieve real-time transportation data. It bridges user interactions with the core ticketing and management systems, ensuring secure handling of customer information and financial transactions while maintaining accessibility and service availability for the public.

2.5.1 Logging in via SSH

The following information applies only to IT Administrators.

1. A secure connection to the Management server can be established by the SSH command with your username, e.g "username@mgmt.team9.isucdc.com". An example is shown in Figure 24.

```
C:\Users\olt_v>ssh melissa.robles@www.team9.isucdc.com
```

Figure 24 SSH command for user 'melisa.robles'

2. After executing the command, the user will be prompted to provide a password. Upon the event of the successful authentication, the user will get a confirmation message as shown in Figure 25. You are now logged into your account and can continue with your tasks.

```
ED25519 key fingerprint is SHA256:FiGfghiAdEy00pLc0hBiDlz7gh+GEhSb3CGsRCfy25s.
This host key is known by the following other names/addresses:
    C:\Users\olt_v/.ssh/known_hosts:2: 33.96.50.50
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'www.team9.isucdc.com' (ED25519) to the list of known hosts.
melissa.robles@www.team9.isucdc.com's password:
Last login: Sun Feb 15 10:21:10 2026 from 199.100.16.101
[melissa.robles@www ~]$
```

Figure 25 Successful Login

3. Whenever the user is ready to logout from the session and wants to terminate the connection, they just need to enter 'exit' or 'logout' which are shown in figures 26 and 27 respectively.

```
Last login: Sun Feb 15 10:21:10 2026 from 199.100.16.101
[melissa.robles@www ~]$ exit
logout
Connection to www.team9.isucdc.com closed.
```

Figure 26 Terminating SSH connection via 'exit' command

```
Last login: Fri Feb 20 22:11:34 2026 from 199.100.16.101
[melissa.robles@www ~]$ logout
Connection to www.team9.isucdc.com closed.
```

Figure 27 Terminating SSH connection via 'logout' command

2.5.2 Logging in Procedure for Public Website

1. Navigate to www.team9.isucdc.com in the Browser as shown in Figure 28

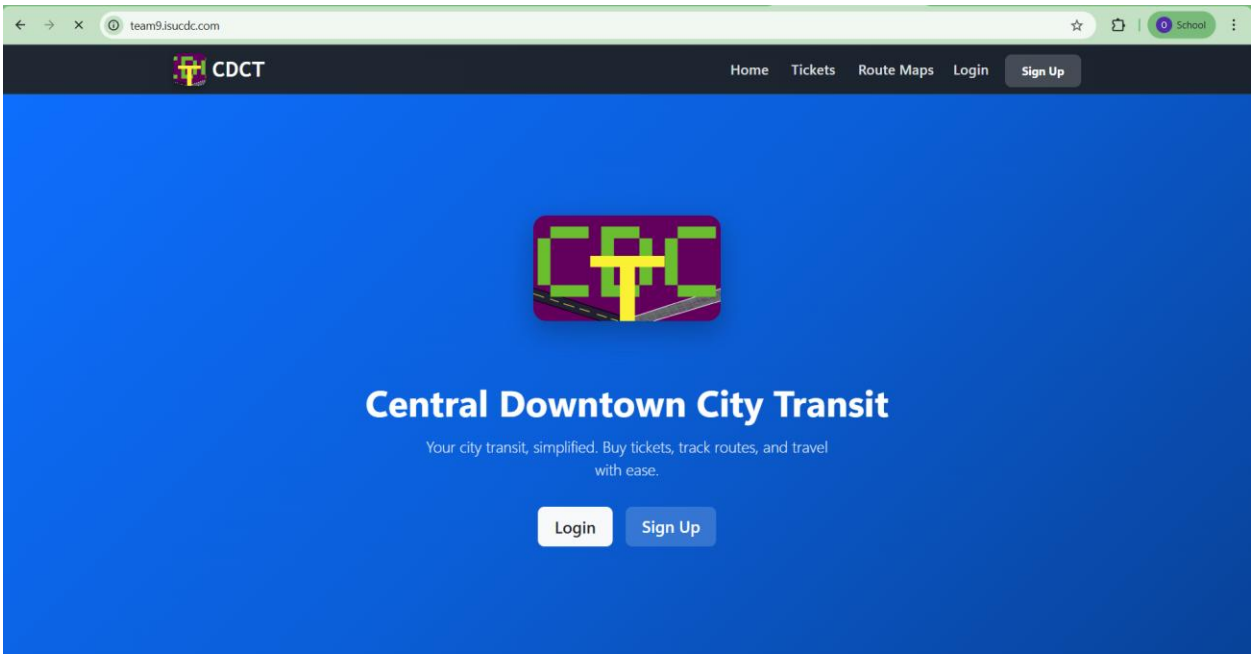

Figure 28 Website Home Page

2. Click on Login and the user will be redirected to the login page, as shown in Figure 29.

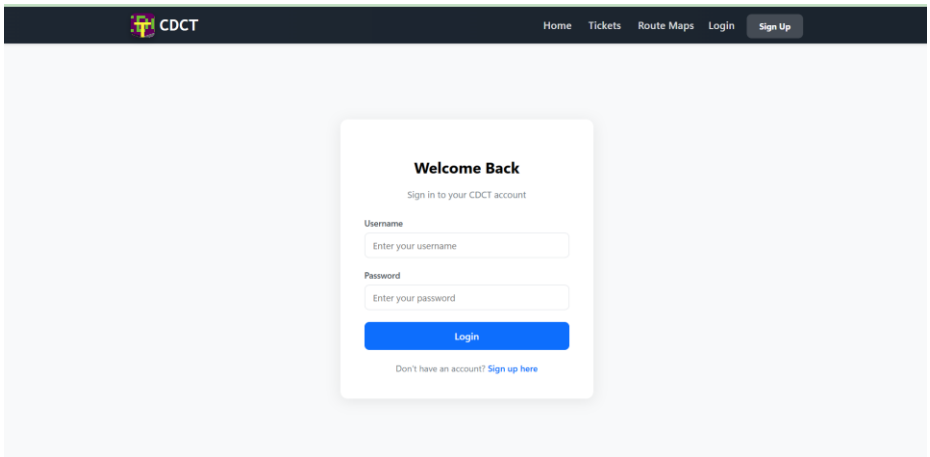

Figure 29 Login Page

3. Authenticate using username and password and the web will redirect you to the page shown in Figure 30. The navbar now contains the users username.

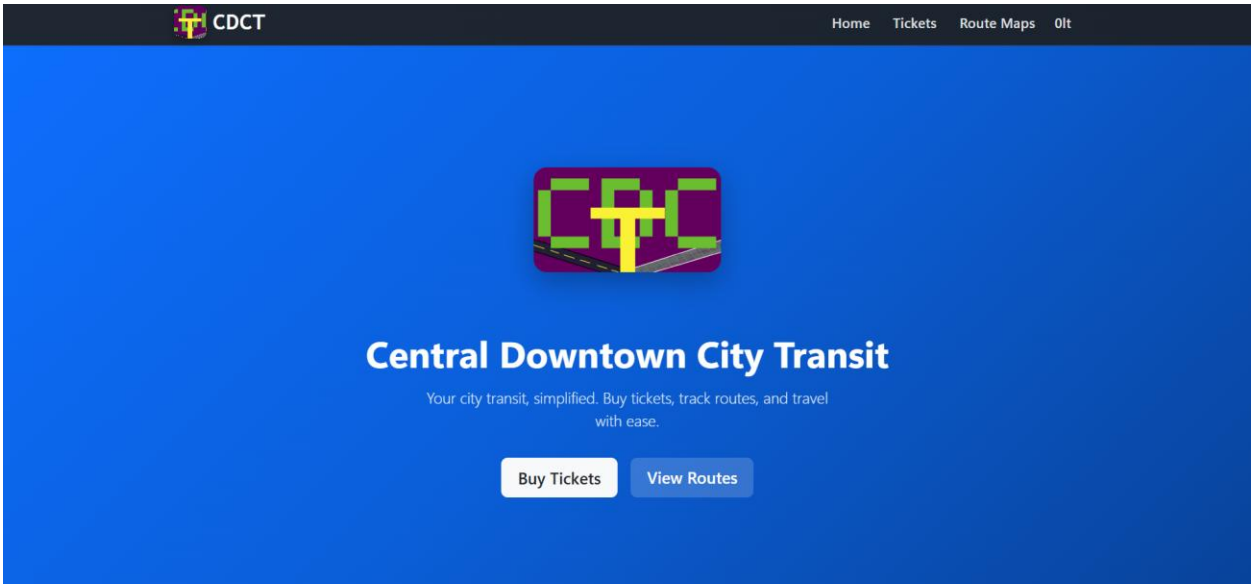

Figure 30 After Successdul Login the Navbar displays the users username

2.5.3 Signing up Procedure for Public Website

1. Navigate to www.team9.isucdc.com in the Browser as shown in Figure 31

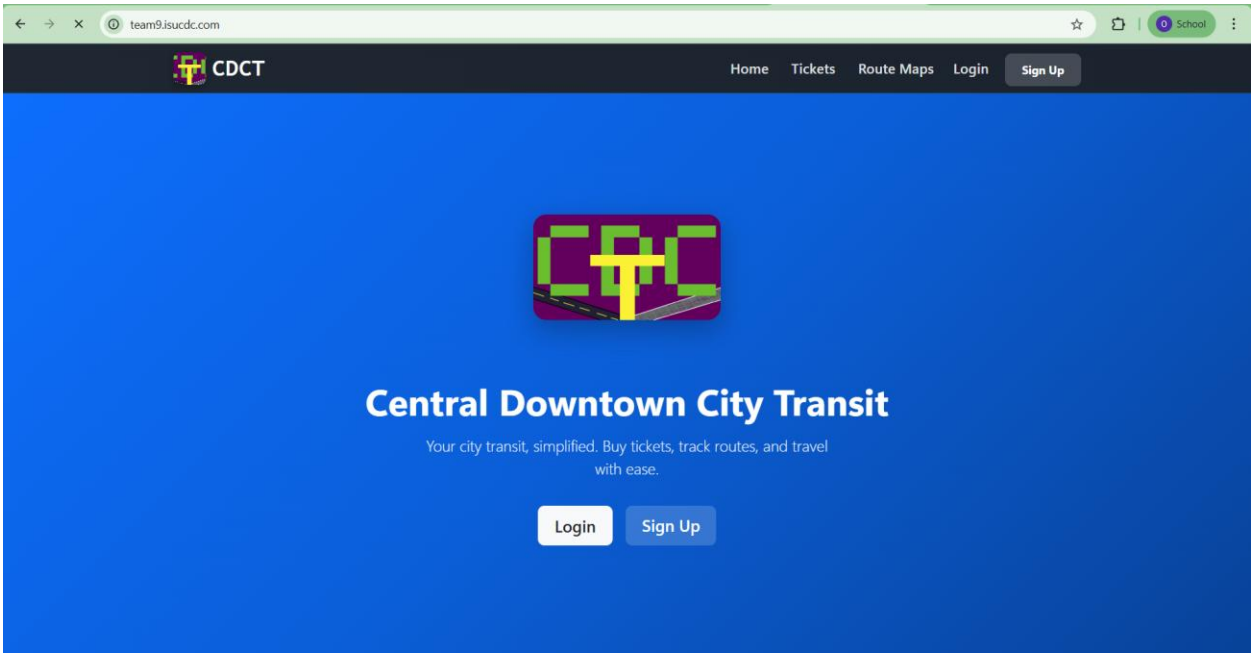

Figure 31 Home Page

2. Click on Sign Up and the user will be redirected to the register page, as shown in Figure 32

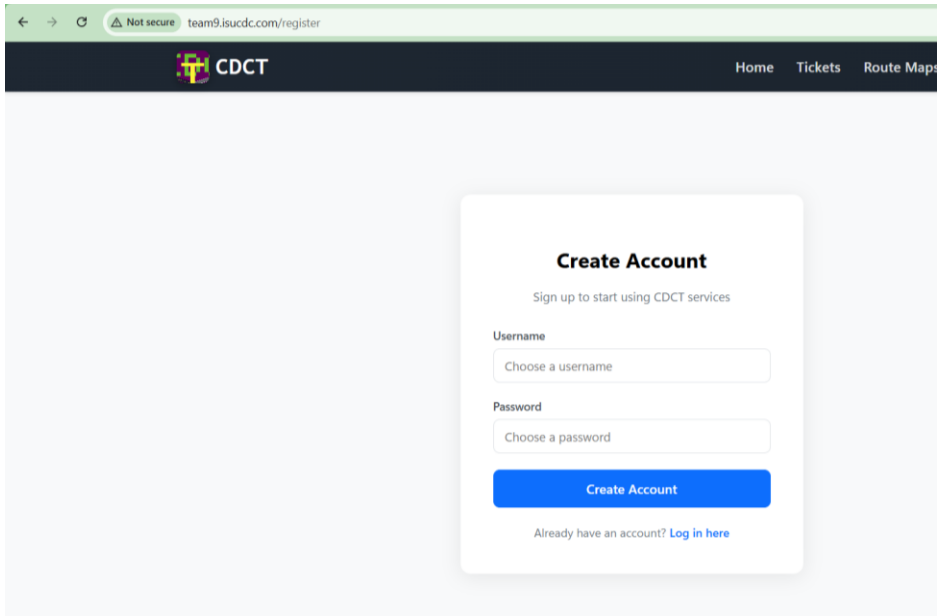

Figure 32 Sign Up Page

3. Fill all required information and click on Create Account. A pop up, like the one in Figure 33,

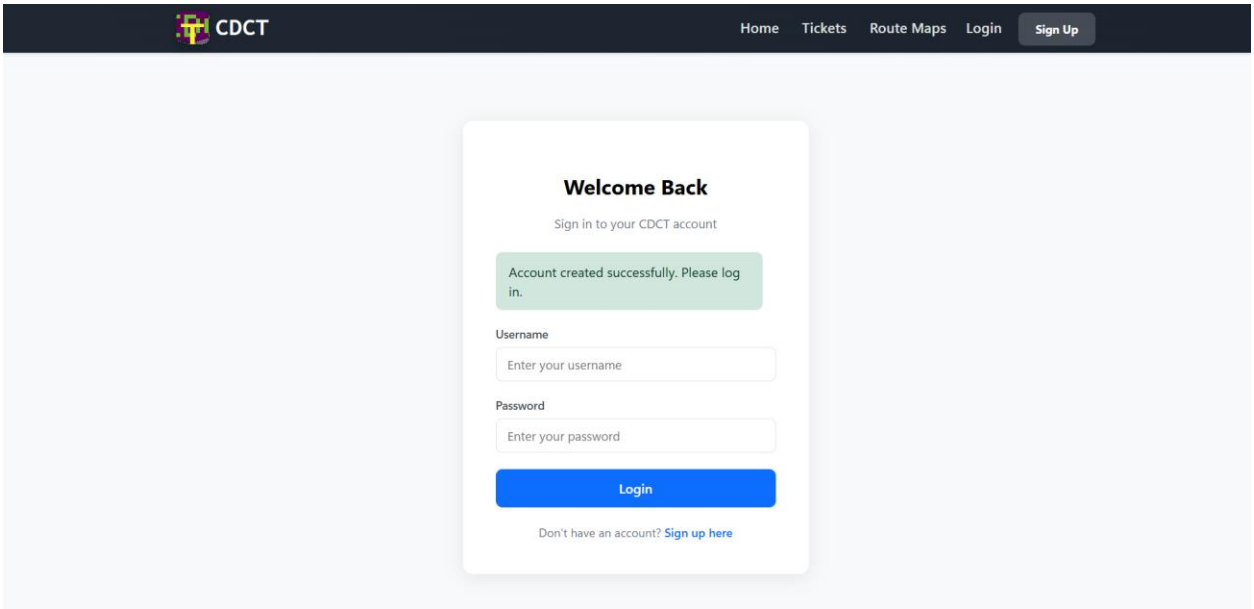

Figure 33 User Created Successfully

2.5.4 Adding Payment Information

1. Authenticate via the login process.

2.Click on the username in the navbar. The users will be redirected to the profile page as shown in Figure 34

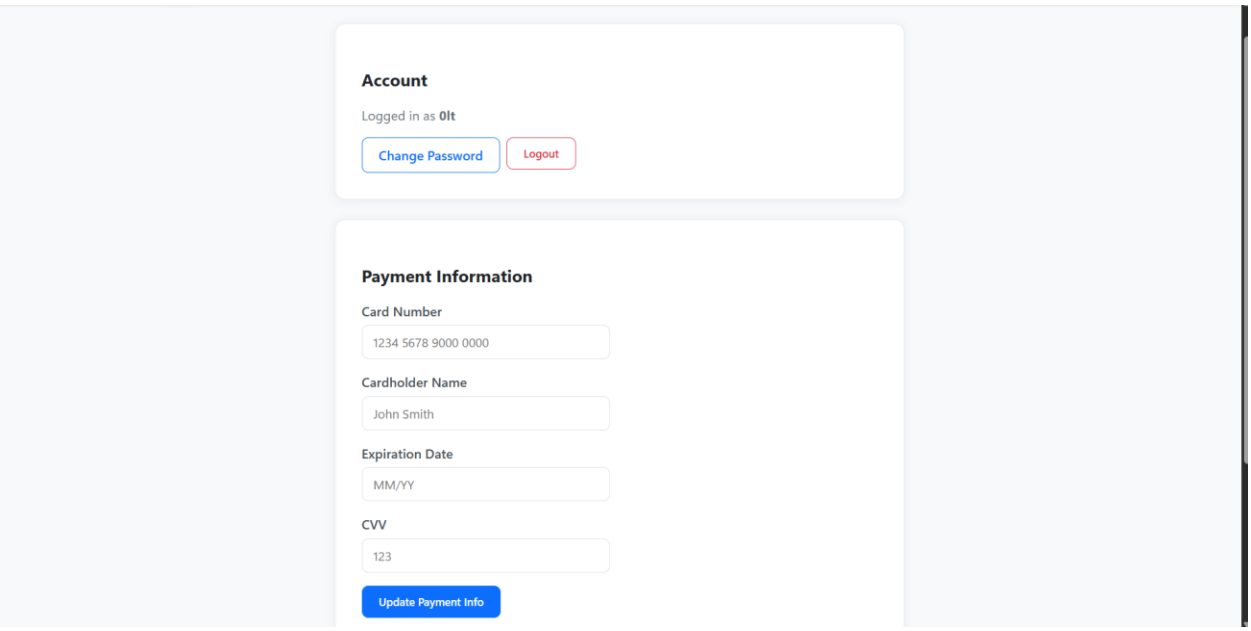

Figure 34 Account Page

3. Fill in the required fields in the payment information.

4. Click on the "Update Payment Info"

5. The page will reload an will show the updated payment info and the announcement on the page will show as demonstrated in Figures 35 and 36

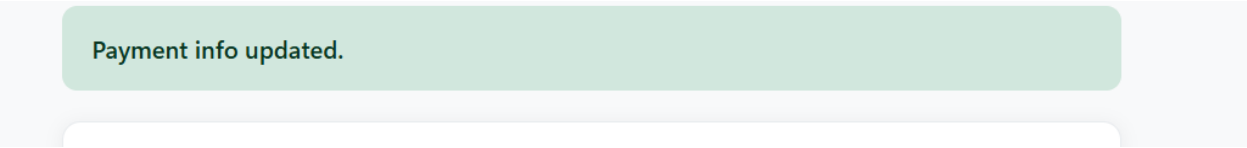

Figure 35 Update Announcement

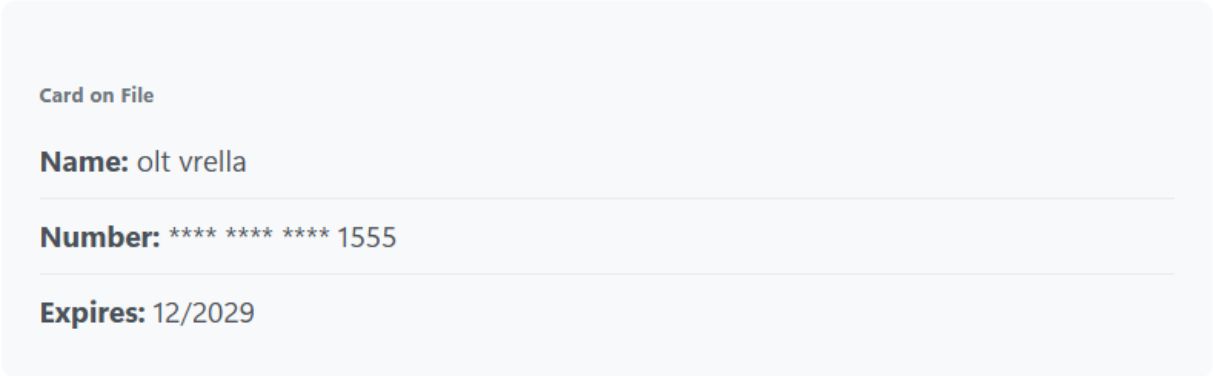

Figure 36 Updated Payment Information

2.5.5 Purchasing and Using a Ticket

1. Authenticate during the login Process and fill in the payment information.

2. Click on Buy Tickets Button on the Home Page or Tickets on the Navbar.

3. The user will be redirected to the Tickets page shown in Figure 37

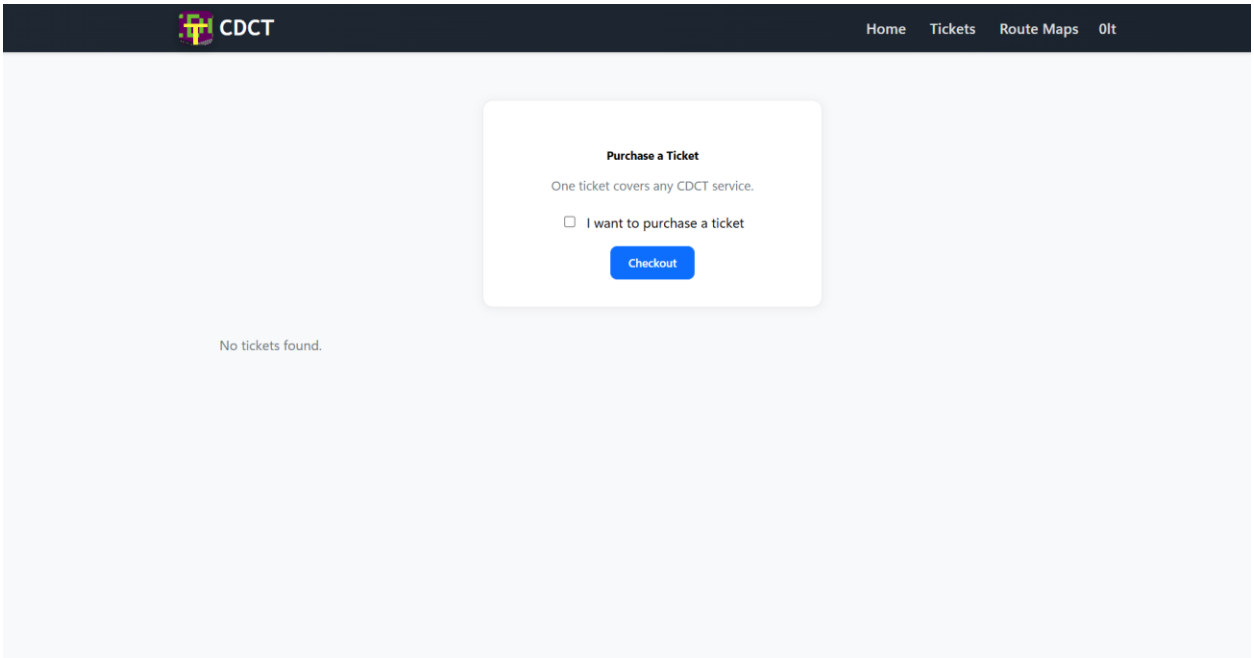

Figure 37 Tickets Page

4. Check the checkbox "I want to purchase a Ticket" and then click the Checkout button

5. After that the ticket will show on the page as demonstrated in figure 38

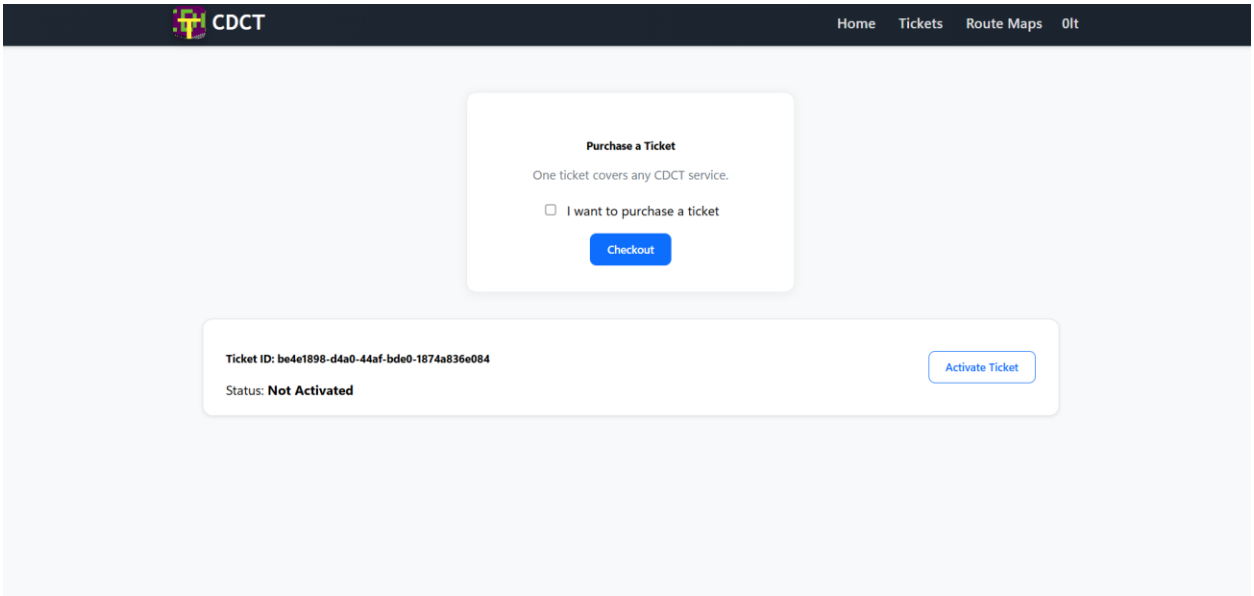

Figure 38 Successful Ticket Purchase

6. Activate the ticket by clicking on the "Activate Ticket" button and a pop up will appear as illustrated in figure 39

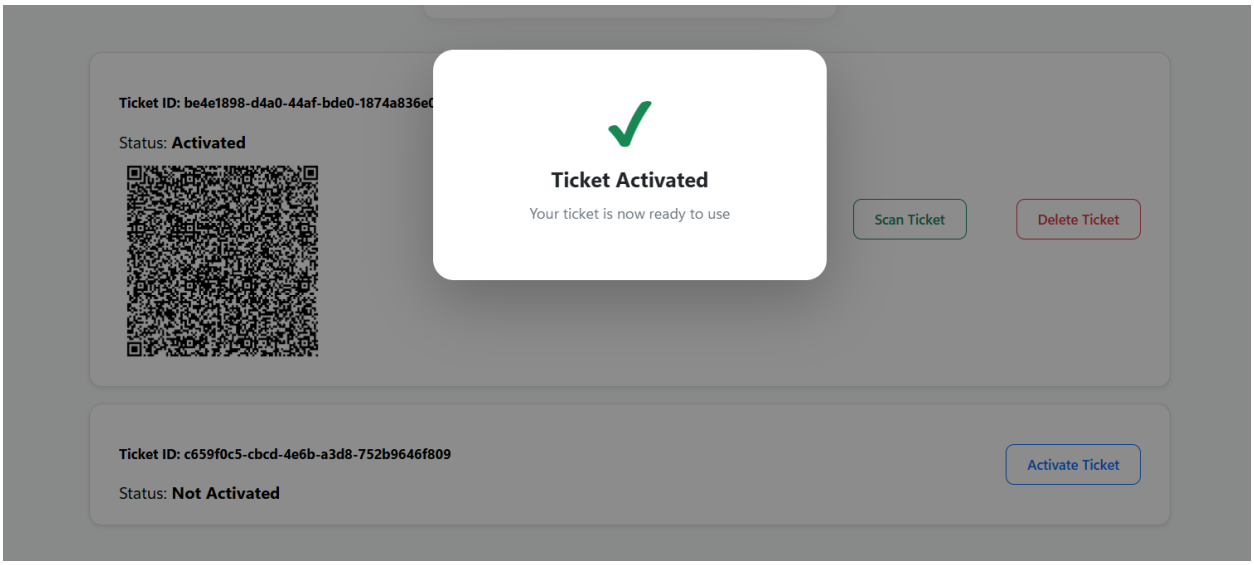

Figure 39 Successful Ticket Activation

7.1 The user can scan the ticket by clicking on the "Scan Ticket" button. A pop up will appear as shown in figure 40

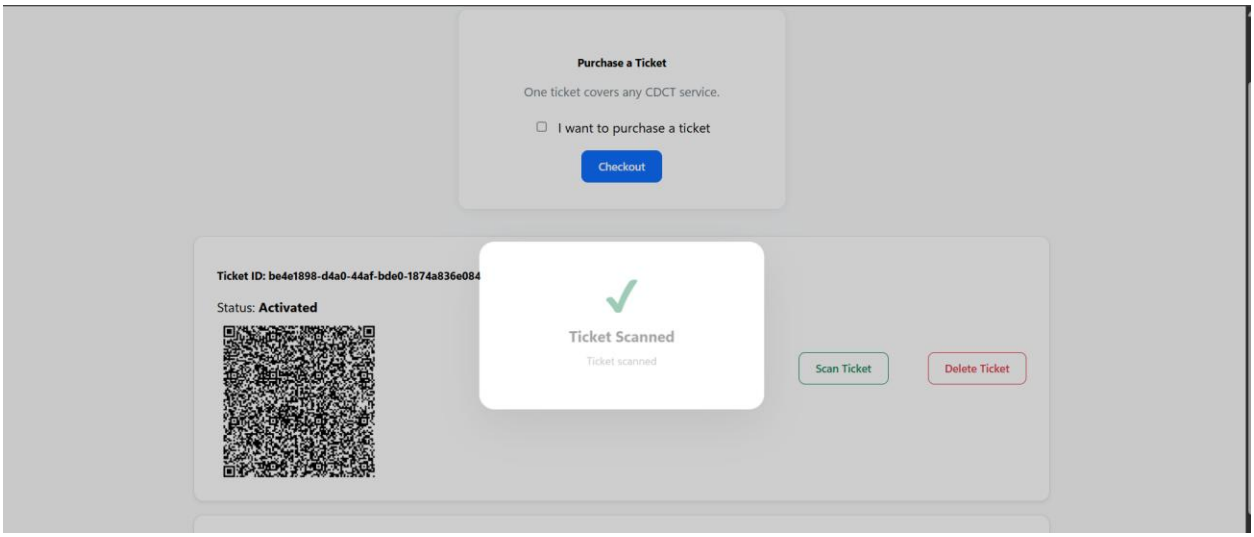

Figure 40 Successful Ticket Scan

7.2 The user can delete the ticket by clicking on the "Delete Ticket" button. A pop up will appear as demonstrated on Figure 41.

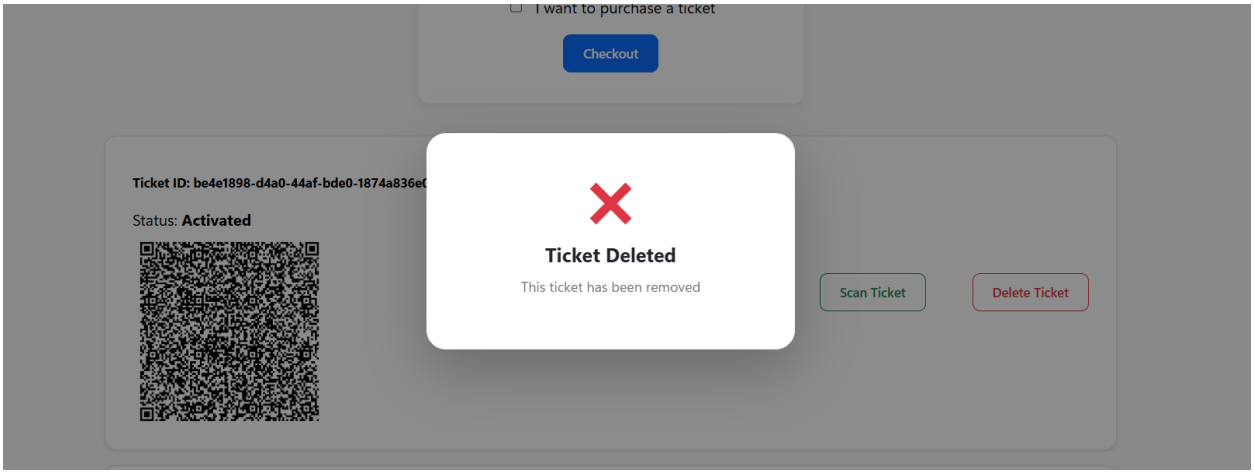

Figure 41 Successful Ticket Deletion

2.5.6 Password Change Process

1. Authenticate via the login process

2. Navigate to the Profile page

3. Click on the "Change Password" button. The users will be redirected to the page shown in Figure 42

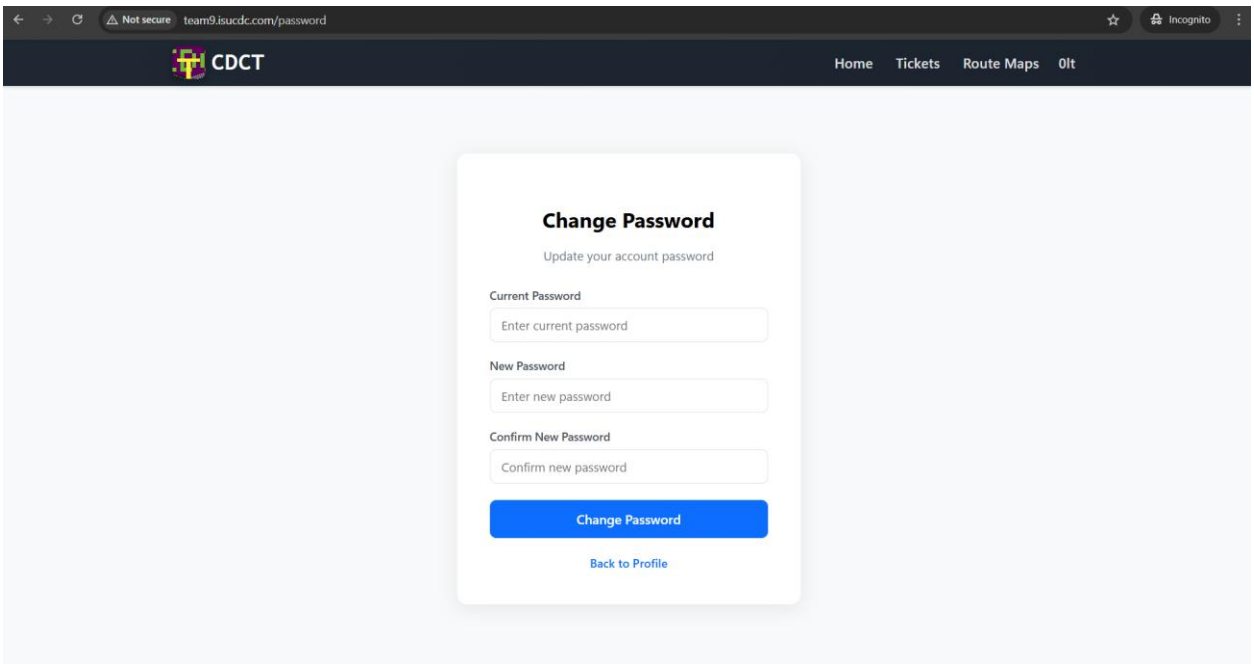

Figure 42 Password Change Page

4. Fill in the information required and click on "Change Password".

5. Observe that after the action has taken place an announcement appears like in Figure 43

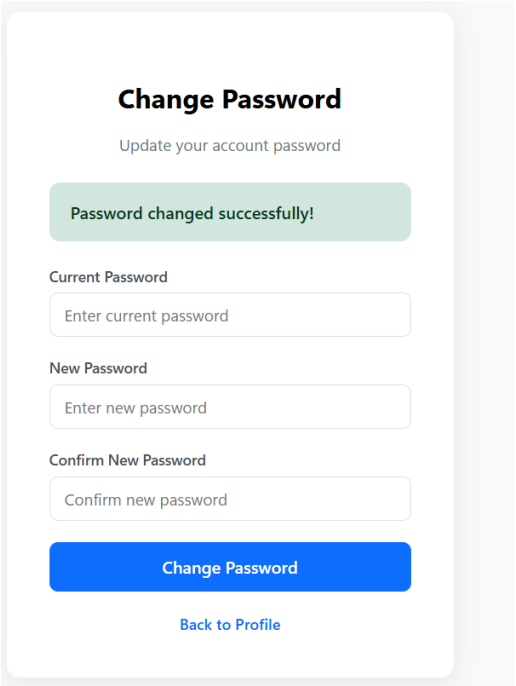

Figure 43 Successful Password Change

2.5.7 Checking the vehicle route map

Note: This can also be done unauthenticated.

1. Navigate to the Public Website
2. Click on the "Route Map" entry on the navbar.
3. The users will be redirected to the maps page as shown in Figure 44

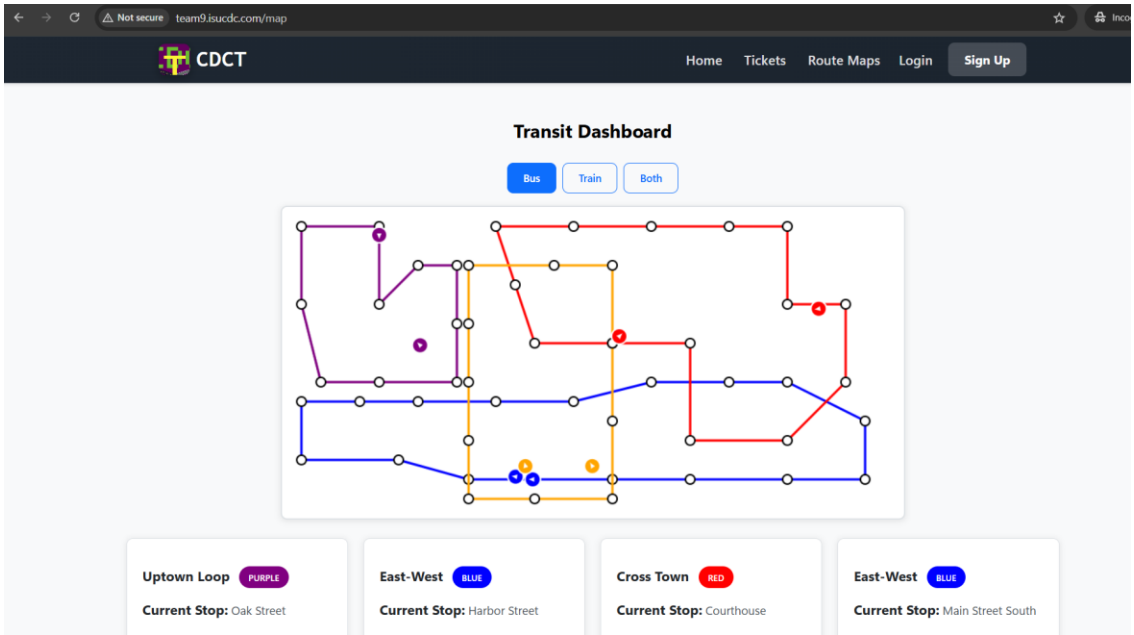

Figure 44 Route Map Page

Note:

The authentication and vehicle endpoints communicate with the Management Backend Server. So every Api function of the Public Website page will function correctly if the Management Backend server is active.

3. References

1. Deland-Han. (2025, 01 15). Understanding the Remote Desktop Protocol (RDP). Retrieved 02 12, 2026, from Microsoft Learn: https://learn.microsoft.com/enus/troubleshoot/windows-server/remote/understanding-remote-desktop-protocol.